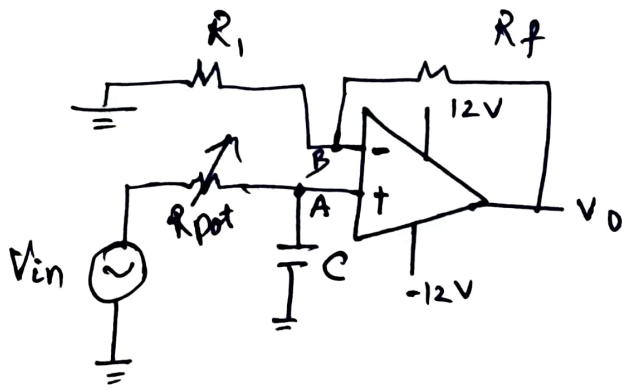


Derivation (First Order - Low pass Butterworth)



Apply KVL.

$$-V_{in} + I(R_{pot}) + I \times C = 0.$$

$$V_{in} = I [R_{pot} + X_C]$$

$$V_A = I \times X_C.$$

And

$$V_d = V^+ - V^-$$

$$A = \left(1 + \frac{R_f}{R_1}\right) V_d$$

$$V_o = \left(1 + \frac{R_f}{R_1}\right) V_A$$

$$V_o = \left(1 + \frac{R_f}{R_1}\right) \frac{V_{in}}{(R_{pot} \omega C(j+1))}$$

$$A' = \frac{\left(1 + \frac{R_f}{R_1}\right)}{(R_{pot} j \omega C + 1)}$$

$$H(s) = \frac{1 + \frac{R_f}{R_1}}{(R_{pot} s C + 1)}$$

Assume

$$R_1 = 1k\Omega \quad R_f = 9k\Omega \quad R_{pot} = 10k\Omega$$

$$C = 0.01\mu F$$